

weighted sums

$$z = w_1x_1 + w_2x_2 + b$$

batch of 4:

2, 4, 6, 8

done separately
for every node
of the layer

1. normalise

$$\mu_B = 5, \sigma_B^2 = 5$$

$$\hat{z} = \frac{z - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}$$

-1.34, -0.45, 0.45, 1.34

mean 0, std 1
(per node,
per batch)

2. scale and shift

$$\gamma = 1.5, \beta = 0.5$$

$$z_{BN} = \gamma\hat{z} + \beta$$

-1.51, -0.17, 1.17, 2.51

γ, β learned
like weights

activation

$$a = g(z_{BN})$$